

VINCE ONG

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Technical Skills

Programming Languages: C++, Python, TypeScript, JavaScript, CSS, SQL, MATLAB

Software & Tools: Git, GitHub, Supabase, Vercel, Arduino IDE, LTspice, EAGLE, backboard.io

Hardware: PCB Assembly, Microcontrollers, Soldering, Oscilloscopes, Function Generators, DMM, 3D Printing

Projects

Engram ↗ | *TypeScript, Python, Supabase*

April 2026 – Present

End-to-End Developer | Althra x TechTO Hacks Finalist; Best use of backboard.io

- Deployed a B2B AI SaaS platform to unify fragmented enterprise workflow data, **reducing employee onboarding time by 30%** through the seamless integration of LLM-powered persistent memory architectures.
- Engineered a scalable semantic data pipeline utilizing Supabase and backboard.io, successfully **indexing over 10,000 records** into interconnected knowledge graphs to enable multimodal voice and text search.
- Architected a custom RAG-driven search interface built in TypeScript, optimizing context-aware retrieval logic across multi-source knowledge bases to consistently deliver **sub-200ms query responses**.

Embedded Circadian Telemetry System ↗ | *C++, I2C, EEPROM*

May 2025 – Present

Independent Embedded Systems Project

- Engineered a state-driven embedded C++ firmware architecture, utilizing **non-blocking timing arrays** to concurrently process asynchronous sensor polling and hardware interrupts without halting continuous main loop execution.
- Integrated diverse hardware peripherals across a shared I2C bus, developing a **non-linear PWM algorithm** utilizing floating-point cosine math to precisely modulate solid-state transistors and simulate natural sunrise illuminance.

Algorithmic Web Runner ↗ | *HTML5, JavaScript, CSS*

April 2026

Front-End Developer | MLH Hack Week Winner

- Built a procedurally generated infinite-scrolling web application in HTML5 and JavaScript, sustaining a **flawless 60 FPS rendering baseline** by implementing custom bounding-box collision detection algorithms.
- Optimized complex asynchronous state management architecture to dynamically track procedural enemy generation, reliably supporting **over 100 concurrent users** with a **P95 latency consistently under 50ms**.
- Integrated a dynamic algorithmic audio engine utilizing object-oriented programming principles, seamlessly modulating complex digital soundscapes in real-time based directly on continuous user input events.

Experience

Resilient Privacy

July 2026 – Present

Software Engineer Intern

Dallas, TX (Remote)

- Architected secure, asynchronous threat intelligence pipelines in **Python** and **Rust**, streamlining the ingestion and analysis of real-world security event data for the KANSHI SaaS cybersecurity platform.
- Engineered scalable backend **APIs** and optimized relational **SQL** database schemas, implementing strategic indexing to ensure robust data delivery and high-performance querying for a React frontend.
- Fortified production infrastructure by authoring rigorous unit and integration tests, and championed professional **Git** workflows to guarantee secure-by-default system deployments.

SFU Racerbot (Autonomous Racing Team)

June 2026 – Present

Software Developer

Burnaby, B.C.

- Contributed on a C++/Python sensor fusion pipeline using **ROS2** to combine 2D LIDAR and camera input for obstacle detection.
- Validated autonomous control algorithms in the **F1TENTH Gym simulator**, streamlining physical testing and mitigating safety-critical bugs prior to hardware deployment.

TELUS Digital AI

June 2026 – Present

Online Data Analyst

Remote

- Contributed to the training data integrity of real-world **AI/ML mapping** systems through labeling and evaluation of large-scale geographical datasets.
- Cross-referenced independent data sources to verify location attributes, resolving discrepancies and flagging edge-case anomalies for downstream AI pipelines.
- Conducted auditory evaluations of text-to-speech (TTS) outputs, assessing delivery, emotional inflection, and phonetic accuracy to refine conversational naturalness of generative audio models.

Education

Simon Fraser University

September 2025 – Present

Bachelor of Applied Science in Mechatronic Systems Engineering – Cumulative GPA: 3.7/4.3

Burnaby, B.C.

- BC Achievement Scholarship | *Awarded to the top 3% of BC Graduates*

October 2025

- International Baccalaureate Diploma | *Graduated with an IB 35/45*

July 2025